Linus Torvalds

A Comprehensive Research Biography

14 June 2026

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Introduction

Linus Benedict Torvalds (born 28 December 1969) is a Finnish and American software engineer who is the creator and lead developer of the Linux kernel since 1991. He also created the distributed version control system Git. Torvalds was one of the recipients of the 2012 Millennium Technology Prize "in recognition of his creation of a new open source operating system for computers leading to the widely used Linux kernel". He is also the recipient of the 2014 IEEE Computer Society Computer Pioneer Award and the 2018 IEEE Masaru Ibuka Consumer Electronics Award.

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Life and career

Early years

Torvalds was born in Helsinki, Finland, on 28 December 1969, the son of journalists Anna and Nils Torvalds, the grandson of statistician Leo Törnqvist and of poet Ole Torvalds, and the great-grandson of journalist and soldier Toivo Karanko. His parents were campus radicals at the University of Helsinki in the 1960s. His family belongs to the Swedish-speaking minority in Finland. He was named after Linus Pauling, the Nobel Prize-winning American chemist, although in the book Rebel Code: Linux and the Open Source Revolution, he is quoted as saying, "I think I was named equally for Linus the Peanuts cartoon character", noting that this made him "half Nobel Prize–winning chemist and half blanket-carrying cartoon character". His interest in computers began with a VIC-20 at the age of 11 in 1981. He started programming for it in BASIC, then later by directly accessing the 6502 CPU directly in machine code (rather than assembly language). He then purchased a Sinclair QL, which he modified extensively, especially its operating system. "Because it was so hard to get software for it in Finland", he wrote his own assembler and "(in addition to Pac-Man graphics libraries)" for the QL, and a few games. He wrote a Pac-Man clone, Cool Man. Torvalds attended the University of Helsinki from 1988 to 1996, graduating with a master's degree in computer science from the NODES research group. His academic career was interrupted after his first year of study when he joined the Finnish Navy Nyland Brigade in the summer of 1989, selecting the 11-month officer training program to fulfill the mandatory military service of Finland. He gained the rank of second lieutenant, with the role of an artillery observer. In 1990, Torvalds resumed his university studies, and was exposed to Unix for the first time in the form of a DEC MicroVAX running ULTRIX. Later, he bought computer science professor Andrew Tanenbaum's book Operating Systems: Design and Implementation, in which Tanenbaum describes MINIX, an educational stripped-down version of Unix. On 5 January 1991 he purchased an Intel 80386-based IBM PC clone before receiving a copy of MINIX, which in turn enabled him to begin work on Linux.

Linux

The first Linux prototypes were publicly released on the Internet in late 1991 from an FTP server at his university. Version 1.0 was released on 14 March 1994. Torvalds first encountered the GNU Project in the autumn of 1991 when another Swedish-speaking computer science student, Lars Wirzenius, took him to the Helsinki University of Technology to listen to free-software guru Richard Stallman's speech. Because of the talk and pressure from other contributors, Torvalds would ultimately switch his original license (which forbade commercial use) to Stallman's GNU General Public License version 2 (GPLv2) for his Linux kernel. After a visit to Transmeta in late 1996, Torvalds accepted a position at the company in California, where he worked from February 1997 to June 2003. He then moved to the Open Source Development Labs, which has since merged with the Free Standards Group to become the Linux Foundation, under whose auspices he continues to work. In June 2004, Torvalds and his family moved to Dunthorpe, Oregon to be closer to the OSDL's headquarters in Beaverton. From 1997 to 1999, he was involved in 86open, helping select the standard binary format for Linux and Unix. In 1999, he was named by the MIT Technology Review TR100 as one of the world's top 100 innovators under age 35. In 1999, Red Hat and VA Linux, both leading developers of Linux-based software, presented Torvalds with stock options in gratitude for his creation. That year both companies went public and Torvalds's share value briefly shot up to about US$20 million. His personal mascot is a penguin nicknamed Tux, which has been widely adopted by the Linux community as the Linux kernel's mascot. Although Torvalds believes "open source is the only right way to do software", he also has said that he uses the "best tool for the job", even if that includes proprietary software. He was criticized for his use and alleged advocacy of the proprietary BitKeeper software for version control in the Linux kernel. He subsequently wrote a free-software replacement for it called Git. In 2000, the CEO of Apple Steve Jobs, proposed to Linus to work on macOS, requiring him to stop his work on the Linux kernel. Linus refused, also arguing that the Mach kernel was too different from Linux. In 2008, Torvalds stated that he used the Fedora Linux distribution because it had fairly good support for the PowerPC processor architecture, which he favored at the time. He confirmed this in a 2012 interview. Torvalds abandoned GNOME for a while after the release of GNOME 3.0, saying, "The developers have apparently decided that it's 'too complicated' to actually do real work on your desktop, and have decided to make it really annoying to do". He then switched to Xfce. In 2013, Torvalds resumed using GNOME, noting that "they have extensions now that are still much too hard to find; but with extensions you can make your desktop look almost as good as it used to look two years ago". The Linux Foundation currently sponsors Torvalds so he can work full-time on improving Linux. In 2012, while giving a talk at Aalto University, Torvalds said "fuck you" and raised his middle finger after criticizing the company Nvidia, which specializes in GPU technology. He said Nvidia was, at the time, the single worst company he has dealt with in the development of the kernel. In the talk, he also discussed other elements of computing. Torvalds is known for vocally disagreeing with other developers on the Linux kernel mailing list. Calling himself a "really unpleasant person", he explained, "I'd like to be a nice person and curse less and encourage people to grow rather than telling them they are idiots. I'm sorry—I tried, it's just not in me." His attitude, which he considers necessary for making his points clear, has drawn criticism from Intel programmer Sage Sharp and systemd developer Lennart Poettering, among others. On Sunday, 16 September 2018, the Linux kernel Code of Conflict was suddenly replaced by a new Code of Conduct based on the Contributor Covenant. Shortly thereafter, in the release notes for Linux 4.19-rc4, Torvalds apologized for his behavior, calling his personal attacks of the past "unprofessional and uncalled for" and announced a period of "time off" to "get some assistance on how to understand people's emotions and respond appropriately". It soon transpired that these events followed The New Yorker approaching Torvalds with a series of questions critical of his conduct. Following the release of Linux 4.19 on 22 October 2018, Torvalds returned to maintaining the kernel. In 2024, amidst the Russian invasion of Ukraine, some developers were excluded from the list of Linux kernel maintainers, seemingly over being Russian or using Russian email addresses. Torvalds commented: "I'm Finnish. Did you think I'd be supporting Russian aggression?" Some developers, along with a part of Linux users, noted the lack of public clarity about that move. Later Torvalds claimed that he acted according to government compliance requirements and due to legal issues around Russia.

Authority and trademark

As of 2006, approximately 2% of the Linux kernel was written by Torvalds. Despite the thousands who have contributed to it, his percentage is still one of the largest. However, he said in 2012 that his own personal contribution is now mostly merging code written by others, with little programming. He retains the highest authority to decide which new code is incorporated into the standard Linux kernel. Torvalds holds the Linux trademark and monitors its use, chiefly through the Linux Mark Institute.

Other software

Git

On 3 April 2005, Torvalds began development on Git, version control software that later became widely used. Torvalds turned it over to Junio Hamano, who became its head maintainer in mid 2005.

Subsurface

Subsurface is software for logging and planning scuba dives, which Torvalds began developing in late 2011. It is free and open source software distributed under the terms of the GNU General Public License version 2. Dirk Hohndel became its head maintainer in late 2012.

Sparse

Sparse is a static analysis tool that flags constructs that are likely to be of interest to kernel developers, such as the mixing of pointers to user and kernel address spaces.

Personal life

Linus Torvalds is married to Tove Torvalds (née Monni), a six-time Finnish national karate champion, whom he met in late 1993. He was running introductory computer laboratory exercises for students and instructed the course attendees to send him an e-mail as a test, to which Tove responded with an e-mail asking for a date. They were later married and have three daughters, two of whom were born in the United States. The Linux kernel's reboot system call accepts their dates of birth (written in hexadecimal) as magic values. Torvalds has described himself as "completely a-religious—atheist", adding, "I find that people seem to think religion brings morals and appreciation of nature. I actually think it detracts from both. It gives people the excuse to say, 'Oh, nature was just created,' and so the act of creation is seen to be something miraculous. I appreciate the fact that, 'Wow, it's incredible that something like this could have happened in the first place.'" He later added that while in Europe religion is mostly a personal issue, in the United States it has become very politicized. When discussing the issue of church and state separation, he said, "Yeah, it's kind of ironic that in many European countries, there is actually a kind of legal binding between the state and the state religion." In "Linus the Liberator", a story about the March LinuxWorld Conference, Torvalds says: "There are like two golden rules in life. One is 'Do unto others as you would want them to do unto you.' For some reason, people associate this with Christianity. I'm not a Christian. I'm agnostic. The other rule is 'Be proud of what you do.'" In 2004, Torvalds moved with his family from Silicon Valley to Portland, Oregon. In 2010, Torvalds became a United States citizen and registered to vote in the United States. As of that year, he was unaffiliated with any U.S. political party, saying, "I have way too much personal pride to want to be associated with any of them, quite frankly." Linus developed an interest in scuba diving in the early 2000s and has achieved numerous certifications, leading him to create the Subsurface project.

Media recognition

Time magazine has recognized Torvalds multiple times: In 2000, he was 17th in their Time 100: The Most Important People of the Century poll. In 2004, he was named one of the most influential people in the world by Time magazine. In 2006, the magazine's Europe edition named him one of the revolutionary heroes of the past 60 years. InfoWorld presented him with the 2000 Award for Industry Achievement. In 2005, Torvalds appeared as one of "the best managers" in a survey by BusinessWeek. In 2006, Business 2.0 magazine named him one of "10 people who don't matter" because the growth of Linux has shrunk Torvalds's individual impact. In summer 2004, viewers of YLE (the Finnish Broadcasting Company) placed Torvalds 16th in the network's 100 Greatest Finns. In 2010, as part of a series called The Britannica Guide to the World's Most Influential People, Torvalds was listed among The 100 Most Influential Inventors of All Time (ISBN 9781615300037). On 11 October 2017, the Linux company SUSE made a song titled "Linus Said" about the origin of the Linux kernel.

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See also

Linus's law Tanenbaum–Torvalds debate List of pioneers in computer science

Further reading

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Chapter 4

Recent Developments and Current Work

Why Linux creator Linus Torvalds gets angry hearing "99% of code is AI" - The New Stack

The New Stack | 2026-05-29

We’re so glad you’re here. You can expect all the best TNS content to arrive Monday through Friday to keep you on top of the news and at the top of your game. Check your inbox for a confirmation email where you can adjust your preferences and even join additional groups. Check out the latest featured and trending stories while you wait for your first TNS newsletter. During his keynote at the recent Open Source Summit North America , Linux and Git creator Linus Torvalds did not mince words about those who contend AI will remove the need for human programmers. Instead of AI “replacing” programming, Torvalds told the audience gathered in Minneapolis that AI boosts productivity in the same way that past technological revolutions did. Programmers will ultimately use AI to generate the source code, which compilers then turn into machine code. However, to build serious projects intended to last decades, developers must deeply understand the generated code and the system architecture, not just the prompts, he clarified. “AI is a great new tool, but it’s a tool, and when I see people saying, ‘Hey, 99% of our code is written by AI,’ I literally get angry, because those same people — I can pretty much guarantee — that 100% of their code is written by compilers. But they never say that.” “When I see people saying, ‘Hey, 99% of our code is written by AI, I literally get angry” From the perspective of a long-time open-source maintainer — the first maintainer of Linux, as its creator — Torvalds views AI as a significant, high-productivity tool, similar to the historical shift from machine code to assemblers and compilers. Torvalds dismissed the idea that AI might one day “replace” programming, arguing instead that it boosts productivity, much like past revolutions in software development. Torvalds emphasized that true software engineering requires, in the immediate future, human understanding of the underlying systems, not just the ability to write AI prompts. “I’m 100% convinced that AI is changing programming, but it’s not changing the fun.” Programmers will ultimately use AI to generate the source code, which compilers then turn into machine code. “I’m 100% convinced that AI is changing programming, but it’s not changing the fun,” Torvalds said. As Torvalds mentioned during his keynote, he initially began programming by working with the bare-number machine language in the pre-assembly days. Then assembly languages came into play, and higher-level languages like Fortran, which used compilers, came along. And then, with compilers, productivity actually increased dramatically for basic programming. AI can increase productivity by a factor of 10, but another way to look at it, according to Torvalds, is that AI-assisted productivity for programming is 10 times less than the productivity gains compilers offer. This assertion is based on Torvalds’ calculation of how compilers have boosted programming by 1,000-fold. “People who don’t understand the complexity of systems will also prompt systems and write processes that will fail. So, I think you do want to understand how it all works.” “People who know what they’re doing to understand systems will be able to prompt tools to write good code,” Torvalds said. “People who don’t understand the complexity of systems will also prompt systems and write processes that will fail. So, I think you do want to understand how it all works.” The immediate dilemma of AI-generated code in open-source projects is the explosion of AI-generated pull requests, often containing bugs that AI tools have found. The people power required to assess and then merge these fixes as commits has been doable for the Linux project for now, thanks to its resources. However, for those thousands of projects that lack resources, many cannot keep up the pace. “Of all the projects that people maintain that are not the Linux kernel, that maybe is somebody’s head project that they’ve been working on for a decade or more, and they have only one person or three people [to patch bugs and create fixes], they get really burned out,” Torvalds said. While AI helps identify deep-seated bugs in aging codebases, it also introduces a social burden: “Drive-by” AI bug reports that lack follow-up, causing maintainer burnout, Torvalds said. Torvalds noted that the Linux project has over 35 years of legacy code and that AI is successfully finding hidden issues. However, it takes extensive time for maintainers to sift through those, Torvalds said. “Sometimes, obviously, AI reports a bug, and when you ask for more information, the person has done that drive-by and doesn’t even answer your question,” Torvalds said. ”So, that’s the real burnout issue.” Additionally, some tech companies invest in making a name for themselves by using AI to flag bugs for media attention — without providing the necessary code patches. “These companies enjoy spending a lot of money and a lot of tokens on pointing out the above, and strangely, none of these came with a patch, even though all of these were in open source code,” Torvalds asserted. “It’s all very good when AI finds a bug in any source code in the short term, but if AI finds issues that we did not find, it’s going to take us some time to just work through the new issues.” “It’s all very good when AI finds a bug in any source code in the short term, but if AI finds issues that we did not find, it’s going to take us some time to just work through the new issues.” The project maintainers saw a surge in pull requests ahead of the 7.1 release, resulting in more commits during preparation than in any previous release. However, it became apparent that the surge in pull requests was due to AI use rather than increased interest in the new release, as Torvalds initially thought. Still, as he noted, contributing to the kernel is a good thing with AI, and that very difficult process has been largely supported and, in many cases, augmented with AI tools. As he noted, there was a roughly 20% increase in submissions overall with the current Linux kernel release. During the Q&A, Torvalds was asked which AI tools the Linux project maintainers use to review pull requests and vulnerability reports, to which he replied, “Sashiko.” However, even with that tool, humans must still maintain and review the implementation or proposed fixes, which still requires a significant amount of manpower across the project. That said, given the number of layoffs in the tech sector, the actual act of programming itself is changing, but humans, at least in the near- to mid-term, will still require significant expertise and remain in demand.

Linus Torvalds to ‘start being more hardnosed’ about ‘pointless pull requests’ – some of which come from AIs - The Register

The Register | 2026-05-24

Warns large release candidates ‘are *not* conducive to long-term stability’ Linux kernel boss Linus Torvalds has signaled he’ll push back when he receives irrelevant pull requests, after complaining that developers are making badly timed and trivial submissions, sometimes after using AI to review code. Torvalds foreshadowed changes in his weekly state of the kernel update , which on Sunday announced the release of a fifth release candidate for version 7.1 of the Linux kernel. “To the surprise of absolutely nobody by now, rc5 is pretty big. Quite a bit bigger than rc5's have traditionally been,” Torvalds wrote, before revealing “I'm not entirely happy about it - most of this is totally trivial stuff to random drivers, which obviously makes it all less scary, but at the same time I'm really not convinced the churn is worth it at rc5 time.” The Linux kernel development cycle usually sees Torvalds open a two-week window during which contributors submit code they hope will make it into the next release. Seven release candidates (rc1-7) follow, with each supposed to represent a step towards delivering a stable update. Revised code always arrives during that process. But high volumes of new contributions to rc5 add complexity at a time work on the new kernel is usually close to completion. “These things are ‘fixes’, sure, but at the same time a lot of them are simply so irrelevant that I think they'd be better off in a linux-next tree and get merged during the merge window,” Torvalds suggested. “And yes, several of these series were triggered by AI code review,” he wrote. “So I think I'll start being a bit more hardnosed about this kind of unnecessary churn this late in the game,” he added. “We are supposed to look for *regressions*. Non-critical fixes to long-standing issues are simply not appropriate for this late in the release cycle.” He then declared rc5 “too big” and said his post is “the heads-up that I'll be pushing back on pointless pull requests with fixes that just aren't that important.” Torvalds justified his new stance on grounds that “these kinds of large rc weeks are *not* conducive to long-term stability.” “Trivial fixes may be trivial, and have a pretty low chance of causing problems, but ‘low chance’ is still not ‘zero chance.’” “Start looking closer at your pull requests, and ask yourself: ‘Is this really a regression or serious enough that it shouldn't just go into the development pile?’” This is the second week in a row that Torvalds has complained that AI is complicating the job of overseeing kernel development, after he last week complained “the continued flood of AI reports has basically made the security list almost entirely unmanageable, with enormous duplication due to different people finding the same things with the same tools.” Minister says trusts can go it alone on procurement as Parliament mulls February 2027 FDP contract renewal

Linus Torvalds says flood of duplicate AI-generated vulnerability reports have made Linux security mailing list 'almost entirely unmanageable' — private list 'a waste of time for everybody involved' in switch to new public system - Tom's Hardware

Tom's Hardware | 2026-05-18

Linus Torvalds declared the Linux kernel's private security mailing list "almost entirely unmanageable" on Sunday in his weekly post to the Linux Kernel Mailing List (LKML), blaming a flood of duplicate vulnerability reports generated by researchers running the same AI tools against the same code. The complaint accompanied the release of Linux 7.1-rc4 and a pointer to newly merged documentation that formalizes how AI-assisted bug reports should be handled. The problem, according to Torvalds, is the combination of volume and redundancy: multiple researchers are independently discovering identical bugs using automated tools and filing them separately on a private mailing list, where nobody can see what has already been submitted. Maintainers end up spending their time triaging duplicates and directing reporters to fixes that were merged weeks earlier. "AI detected bugs are pretty much by definition not secret, and treating them on some private list is a waste of time for everybody involved," Torvalds wrote on LKML. Torvalds pointed developers to the project's security bug documentation, which states that vulnerabilities found using AI tools should be treated as public disclosures and submitted directly to the relevant maintainers, not routed through the private security list. Reports must be concise, formatted in plain text, and include a verified reproducer. In March, Willy Tarreau, the creator of HAProxy and a longtime Linux kernel stable maintainer, said in comments posted to LWN that the kernel security mailing list, which received roughly two to three reports per week two years ago, now receives five to 10 reports per day. Most are solid finds, but the duplication across researchers using similar tooling has overwhelmed the existing triage process. Torvalds urged researchers to go further than filing raw findings. "If you actually want to add value, read the documentation, create a patch too, and add some real value on top of what the AI did," he wrote. "Don't be the drive-by 'send a random report with no real understanding' kind of person." This Torvalds-endorsed approach is exactly what fellow maintainer Greg Kroah-Hartman has been doing with his “Clanker T1000” system , a Framework Desktop-powered bug-finding tool: discover the issue, write the fix, take responsibility for the patch, and submit it publicly. Get Tom's Hardware's best news and in-depth reviews, straight to your inbox. The Linux kernel project formalized its broader stance on AI-assisted contributions last month, establishing a project-wide policy that permits AI-generated code provided developers follow strict disclosure rules. Under that policy, AI agents cannot use the legally binding "Signed-off-by" tag, and contributors must use a new "Assisted-by" tag for transparency. Every line of AI-generated code, and any resulting bugs, remains the legal responsibility of the human who submits it. Follow Tom's Hardware on Google News , or add us as a preferred source , to get our latest news, analysis, & reviews in your feeds. Luke James is a freelance writer and journalist. Although his background is in legal, he has a personal interest in all things tech, especially hardware and microelectronics, and anything regulatory.

Linus Torvalds on the AI claim that makes him angry, and what security researchers should never do - ZDNET

ZDNET | 2026-05-21

Linus Torvalds and Dirk Hohndel at Open Source Summit North America 2026 Speaking at the Linux Foundation's Open Source Summit North America , Linux creator Linus Torvalds said modern AI tools are reshaping how developers work on the kernel, driving up contribution volume and exposing new social and security stresses in the open■source world. But he insisted "AI is a great tool, but it's a tool" rather than a wholesale replacement for programmers. Also: When revealed data brings AI rollouts to a screeching halt - and how to manage it Now, if only the companies laying off tech workers left and right would listen. Torvalds spoke with Verizon's Open Source Program Office head, Dirk Hohndel, who is also a Linux kernel maintainer -- and Torvalds' friend. Torvalds added that while the Linux kernel's long■standing release process has been stable "for pretty much exactly 20 years" since the move to Git, that trend broke about six months ago as AI coding tools took off. "In the last six months, we've seen a lot more commits," Torvalds noted, estimating that "the last two releases, it's been about 20% more commits than we had in the previous releases over many years." Initially, Torvalds misread the spike as excitement around a major version change: "At first I thought, 'hey, people are excited about the 7.0 release because I changed the major number every once in a while…' and it turns out I was wrong. The real change that happened in the last six months was that the AI tools actually got good enough for a lot of people... we're seeing a definite uptick in just development on pretty much all fronts." Also: Ubuntu Core 26 offers an immutable Linux you can trust through 2041 Torvalds acknowledged that the new tools lower the barrier of entry for contributors, echoing Hohndel's observation that "the tooling actually lowers this initial barrier… [and] does a big chunk of the work." But he emphasized that the real impact is social rather than purely technical: "The big pain points in Linux, traditionally, and I suspect in most projects, have not been so much the code itself, but… when you are forced to change how you work." One of the biggest flashpoints has been the Linux kernel security mailing list, which Torvalds said was recently "overrun by duplicate reports" generated with AI. "People think that when they find a bug with AI, the first reaction sometimes seems to be, let's send it to the security list, because this may have security implications," he said. The result, on a deliberately small, confidential list, was that "we were flooded by people sending bugs, and then you have this list with very few people on it… and we spent all our time just forwarding these reports to… the other developers who knew that area better." To cope, Torvalds announced new AI security disclosure guidelines with a blunt rule: "If you find a security bug with AI, you should basically consider it to be public, just because if you found it with AI, 100 other people also found it with AI." At the same time, he urged researchers not to publish working exploits: "When it comes to things that really are security issues, you may not want to make the exploit public… Don't be that guy who then crows about it publicly and says, 'Look, I could bring down this big company.'" Torvalds linked the disclosure debate to broader shifts in the security ecosystem. In the past, he said, the kernel community would quietly notify distributions about a bug and ask them to upgrade without detailing the vulnerability, and "most of the time, nobody would figure out what happened." Now, with AI■accelerated analysis, he recalled that "last week, we fixed the bug; within three hours, there was a blog post about the implications of that bug fix, because security people love getting attention." Also: Open-source security is a mess - IBM and Red Hat bet $5 billion and 20,000 engineers can fix it He went out of his way to argue that closing the source is not an answer: "I don't think, for example, that the solution is to not do open source, because if you think that AI can't reverse engineer closed source, you're in for a surprise." In fact, he warned, "closed source is even worse in this respect, because the AI can't help you fix the problems, but the AI sure can help find those problems in the first place." Torvalds is right. While Windows vulnerabilities, except for the truly horrid ones, no longer receive much attention, AI is also finding plenty of security holes in Windows as well. As Dustin Childs, head of threat awareness at Trend Micro's Zero Day Initiative, observed recently, " Microsoft's total count came to 1,139 CVEs patched in 2025 ," which was the second-highest, behind 2020. Childs expects, "as AI bugs become more prevalent, this number is likely to go higher in 2026." Meanwhile, back at Open Source Summit, Hohndel criticized vendors who hype vulnerabilities without responsibly coordinating fixes. He cited four recent local privilege escalation bugs in the kernel, "two of which were disclosed exactly" with branded names, domains, and logos before maintainers were contacted. "My response is always, here is a company I never want to work with, because if you do that to the Linux kernel, you do this to anyone." As annoying as this is, Torvalds admitted to having a love■hate relationship with AI. "I actually really like it from a technical angle. I love the tools. I find it very useful and interesting, but it is definitely causing pain points," he said. Also: 10 trillion downloads are crushing open-source repositories - here's what they're doing about it On the positive side, he framed AI■discovered bugs as "short-term pain" with long■term benefits: "When AI finds a bug in any source code… long term is you found a bug, we fixed it, that the end result is better for it." After all, he continued, "I think finding bugs is great, because the real problem is all the bugs you didn't find." But he warned of "social choke points and social pain points" as AI pours traffic into already overstretched communities, especially in the "10s of 1,000s of random projects that people maintain that are not the Linux kernel." For small teams or solo maintainers, he said, flood■style AI bug reports can cause real burnout, especially when "it's a bug report, and when you ask for more information, the person has done a drive-by and doesn't even answer your questions anymore." Torvalds added that maintenance is increasingly about people rather than code. "For me, as a top-level maintainer, I don't do a lot of coding. My job is working with people, and I do not use AI to work with people. Thank you. And I should suggest you don't do that either." Torvalds has come a long way from the days when he was known for treating poor coders with contempt. Stepping away from Linux, when asked what advice he would give to someone at the beginning of their career amid doom■and■gloom forecasts that "all code will be written by AI," Torvalds pushed back hard on marketing claims. "My opinion has always been that AI is a great tool, but it's a tool, and when I see people saying, 'hey, 99% of our code is written by AI,' I literally get angry." Also: Microsoft surprises with its first server Linux distribution: Azure Linux 4.0 He contrasted those claims with the reality that "100% of their code is written by compilers," and traced his own path from hand■entered machine code to assemblers, then compilers, and now AI helpers. "I grew up writing machine code, and when I say machine code, I don't mean assembly language, I mean the numbers," he said, recalling that "it took me a while to understand that writing down the numbers and calculating offsets for branches is kind of stupid, and people had come up with this tool called an assembler, and then later on I figured out compilers are good too. These days, I'm figuring out AI tools are good too." So, Torvalds argued, "I'm personally 100% convinced that AI is changing programming, but it's not changing the fundamentals." Just as compilers increased productivity "by a factor of 1000," he estimates that "AI will increase your productivity by a factor of 10," but insists "AI is great, but AI is not changing programming." Instead, he contended, "a lot of people will use AI to generate the code that the compilers use to generate the code that the assemblers then use to generate the machine code. This is revolutionary in the same sense that we've seen revolutions before." Crucially, Torvalds said, would■be developers still need to understand what their tools produce. "You do want to understand how it all works in the end," he said. "Even when I use AI for my pet toy projects, I will use AI to generate code, I will look at that code, I will actually still look at the assembly language… because it's what I grew up with." For any serious, long■lived system, he warned, "you need to understand not just your prompts, but you need to understand the end result

Linus Torvalds is fed up with AI-generated bug reports bloating the Linux kernel - TechSpot

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What just happened? Linus Torvalds has expressed frustration over Linux developers submitting ill-timed bug reports just before an RC5 release, with some using AI to detect trivial issues. He added that several of the fixes are also being written by AI, and that the AI-generated code often adds bloat to the Linux kernel rather than addressing the actual problem. In his weekly state of the kernel update , Torvalds noted that the new RC5 is much larger than any other RC5 in recent memory, and he blamed developers for using AI to submit "pointless pull requests." Torvalds added that he is "not entirely happy" about the additional bloat, especially because the changes can largely be attributed to "totally irrelevant stuff." Torvalds urged developers to be more diligent about their pull requests and warned that going forward, he will have to be more "hard-nosed" about unnecessary churn, especially when it's late in the cycle. According to him, in the days leading up to an RC5 release, developers should look for regressions rather than spending time on non-critical fixes to long-standing issues. Explaining his stance on smaller bugs, Torvalds said that even trivial fixes could introduce new issues that could hurt long-term system stability, so developers must avoid submitting new fixes late in the cycle. "Trivial fixes ... may have a pretty low chance of causing problems, but low chance is still not zero chance," he noted. Torvalds has long been warning about the increasing number of AI-generated bug reports that he claims are breaking Linux kernel development. Following the release of the fourth release candidate of Linux 7.1 earlier this month, he noted that duplicate AI-generated reports are clogging security channels, creating unnecessary work for the few developers who actually try to resolve the problems. Claiming that the security mailing list has become unmanageable due to a sharp rise in AI-powered bug reports, Torvalds noted that developers need to be aware of the "enormous duplication" in bug reports right now, caused by different people finding the same bugs using the same AI tools. It is worth noting that Torvalds is not against vibe coding. Earlier this year, he announced that he had joined a long list of well-known developers who use generative AI to write code. According to the Linux creator, he used Google's Antigravity AI - a coding assistant built to generate code via natural-language input - to generate Python code for a side project called AudioNoise.

References

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